

Prenatal exposure to polychlorinated biphenyls and dioxins from the maternal diet may be associated with immunosuppressive effects that persist int...



We investigated whether prenatal exposure from the maternal diet to the toxicants polychlorinated biphenyls (PCBs) and dioxins is associated with the development of immune-related diseases in childhood. Children participating in BraMat, a sub-cohort of the Norwegian Mother and Child Cohort Study (MoBa), were followed in the three first years of life using annual questionnaires (0-3years; n=162, 2-3years; n=180), and blood parameters were examined at three years of age (n=114). The maternal intake of the toxicants was calculated using a validated food frequency questionnaire from MoBa. Maternal exposure to PCBs and dioxins was found to be associated with an increased risk of wheeze and more frequent upper respiratory tract infections. Furthermore, maternal exposure to PCBs and dioxins was found to be associated with reduced antibody response to a measles vaccine. No associations were found between prenatal exposure and immunophenotype data, allergic sensitization and vaccine-induced antibody responses other than measles. Our results suggest that prenatal dietary exposure to PCBs and dioxins may increase the risk of wheeze and the susceptibility to infectious diseases in early childhood. Copyright © 2012 Elsevier Ltd. All rights reserved.